Power and resource tradeoffs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Pilot experiments. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of power and resource tradeoffs. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **power** and resource tradeoffs.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **power** and resource tradeoffs

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Pilot experiments. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **power** and resource tradeoffs using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Power and resource tradeoffs case. Separate observations, given facts and assumptions before choosing an explanation.

Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Power and resource tradeoffs. Connect each labelled input, deciding step and result.

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Power and resource tradeoffs — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
For a fixed alternative100 simulations detect62 times at n20 and83 at n40. Per-unit cost3 credits.	Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.	Estimated detection62% versus83%; cost60 versus120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Power and resource tradeoffs. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For a fixed alternative100 simulations detect62 times at n20 and83 at n40. Per-unit cost3 credits. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Change the assumed effect or variability but keep the sample sizes.

Predicted result

Prediction: The detection fractions need new simulation; the resource costs stay 40 and 80.

Why the result follows

Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Power and resource tradeoffs to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Power	The test's detection probability under a specified alternative
Simulation trial	A generated dataset under a declared model
Planning effect	The effect size used for design calculations

Use two terms to explain a step in this week’s example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Power and resource tradeoffs, predict a step, then compare with its explanation.

For a fixed alternative100 simulations detect62 times at n20 and83 at n40. Per-unit cost3 credits.

Estimated detection62% versus83%; cost60 versus120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

Worked explanation · Power and resource tradeoffs

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Power and resource tradeoffs

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

The facts we will use

Supplied facts

There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

AI ACADEMY / AI-7/23

Stipulated simulation and resource records

	Independent units	Detections	Trials	Cost units
Smaller design	20	60	100	40
Larger design	40	80	100	80

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/23

Power and resource tradeoffs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Estimate simulated detection

Observed detection fractions are 60/100=60% and 80/100=80%.



2. Calculate design cost

n=20 costs 40 resource units; n=40 costs 80 units.



3. Compare the trade-off

Doubling cost adds 20 percentage points of simulated detection under these assumptions.

CHANGE ONE THING

Change the assumed effect or variability but keep the sample sizes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Estimate simulated detection	Observed detection fractions are 60/100=60% and 80/100=80%.
2. Calculate design cost	n=20 costs 40 resource units; n=40 costs 80 units.
3. Compare the trade-off	Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

What this example does not establish

Keep the stated limit

Only 100 trials per design produce noisy estimates. They are fictional stipulated results, not

executed simulation or **evidence** about pupils.

Keep the stated limit

Read the labelled picture: Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

Keep the stated limit

Change one thing: Change the assumed effect or variability but keep the sample sizes. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Power and resource tradeoffs claim and explain a repair using the supplied evidence.

Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Power and resource tradeoffs with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Power** and resource tradeoffs". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Power** and resource tradeoffs. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Power and resource tradeoffs. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For a fixed alternative100 simulations detect62 times at n20 and83 at n40. Per-unit cost3 credits. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Power and resource tradeoffs. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled **power** and resource tradeoffs **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **power** and resource tradeoffs.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Power and resource tradeoffs case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Power and resource tradeoffs — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
For a fixed alternative100 simulations detect62 times at n20 and83 at n40. Per-unit cost3 credits.	Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.	Estimated detection62% versus83%; cost60 versus120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

One hundred alternative trials detect70 at n30 and85 at n60; unit cost2. Compare.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

One hundred alternative trials detect70 at n30 and85 at n60; unit cost2. Compare.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Power and resource tradeoffs, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

By the end, I can	How I will show it
Explain the mechanism and terminology of power and resource tradeoffs	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Power and resource tradeoffs without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **power** and resource tradeoffs

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Doubling cost adds 20 percentage points of simulated detection under these assumptions.

1. Compare the trade-off

2. Estimate simulated detection

3. Calculate design cost

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare the trade-off Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the assumed effect or variability but keep the sample sizes. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The detection fractions need new simulation; the resource costs stay 40 and 80.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The detection fractions need new simulation; the resource costs stay 40 and 80.

Power depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Repair a missing building block

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

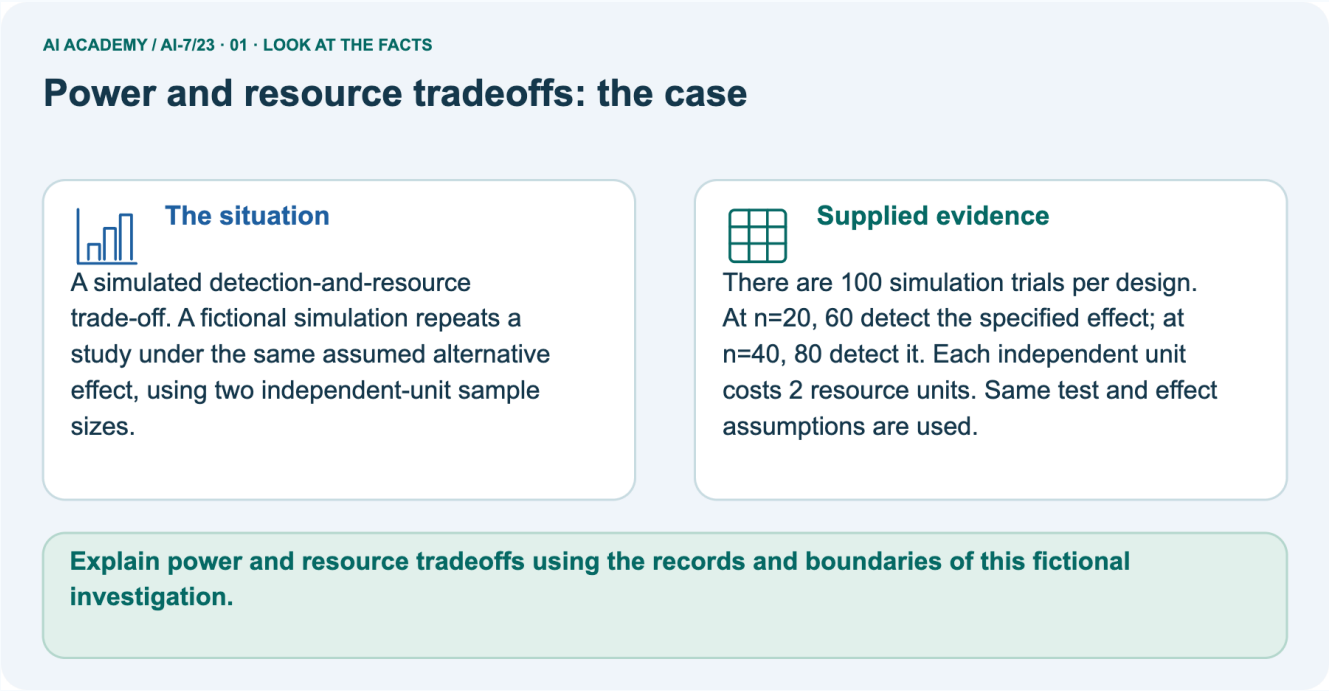
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Power and resource tradeoffs graphic using its labels, deciding evidence and relevant limitation.

01. Power and resource tradeoffs: the case



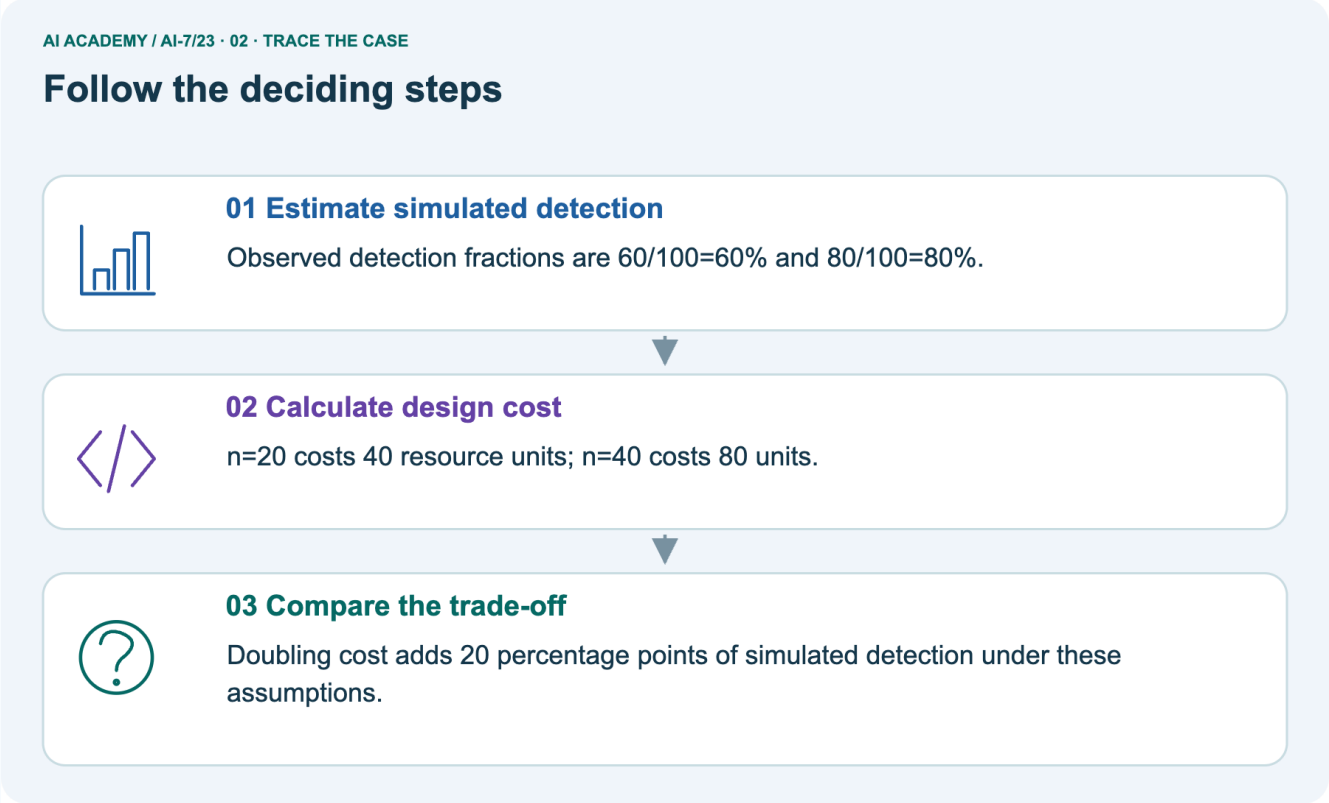
Follow the labelled picture with the matching explanation. Open the vector diagram

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

Supplied facts: There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

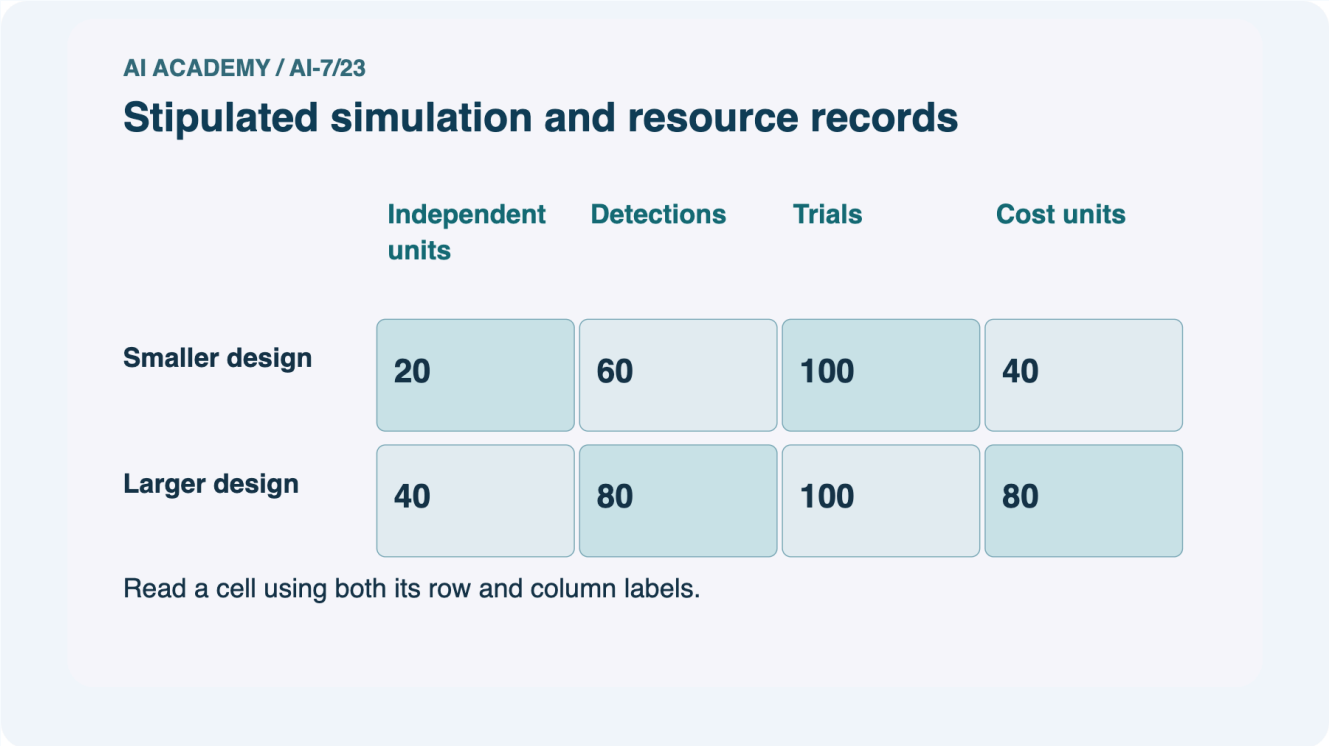


Follow the labelled picture with the matching explanation. Open the vector diagram

Estimate simulated detection: Observed detection fractions are 60/100=**60%** and 80/100=**80%**. → Calculate design cost: n=20 costs 40 resource units; n=40 costs 80 units. → Compare the trade-off: Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Point to each numbered stage and explain which recorded **input** it uses.

03. Stipulated simulation and resource records

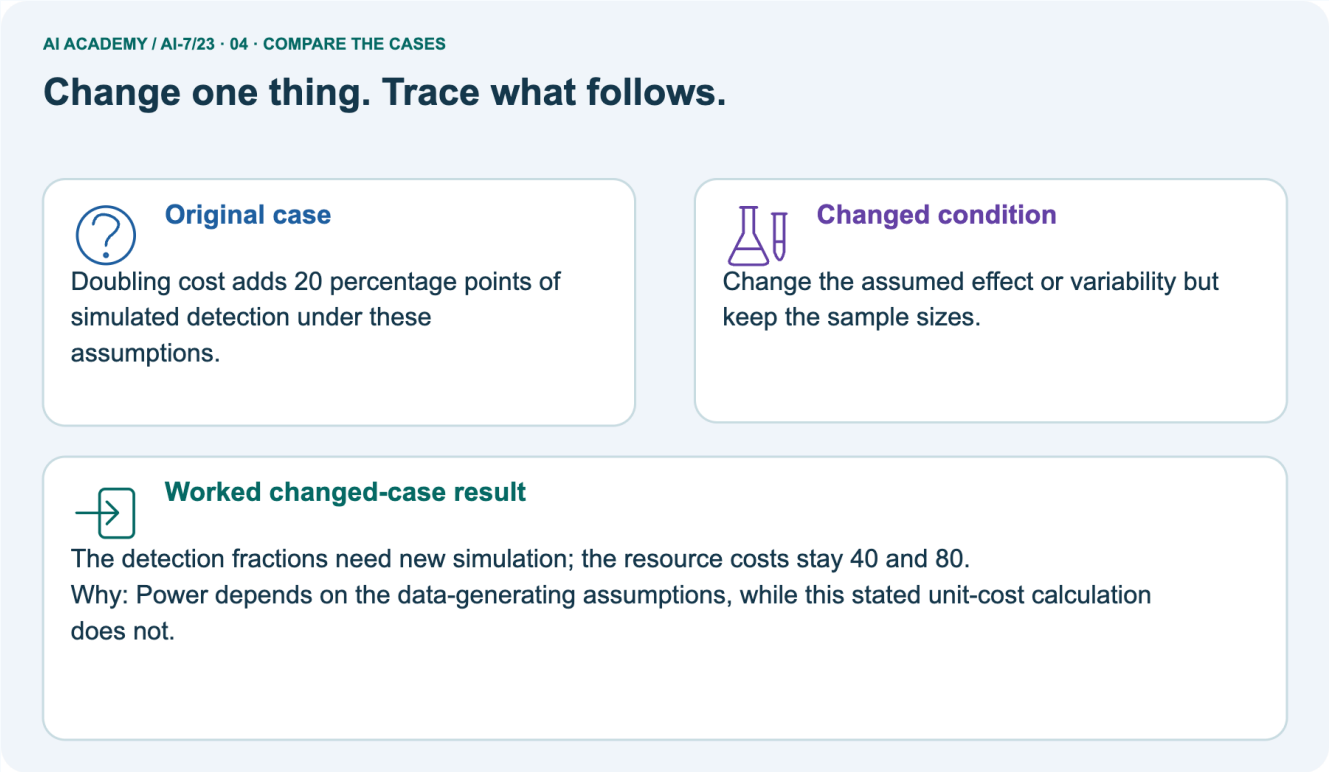


Follow the labelled picture with the matching explanation. Open the vector diagram

Stipulated simulation and resource records. Smaller design: Independent units = 20, Detections = 60, Trials = 100, Cost units = 40. Larger design: Independent units = 40, Detections = 80, Trials = 100, Cost units = 80.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



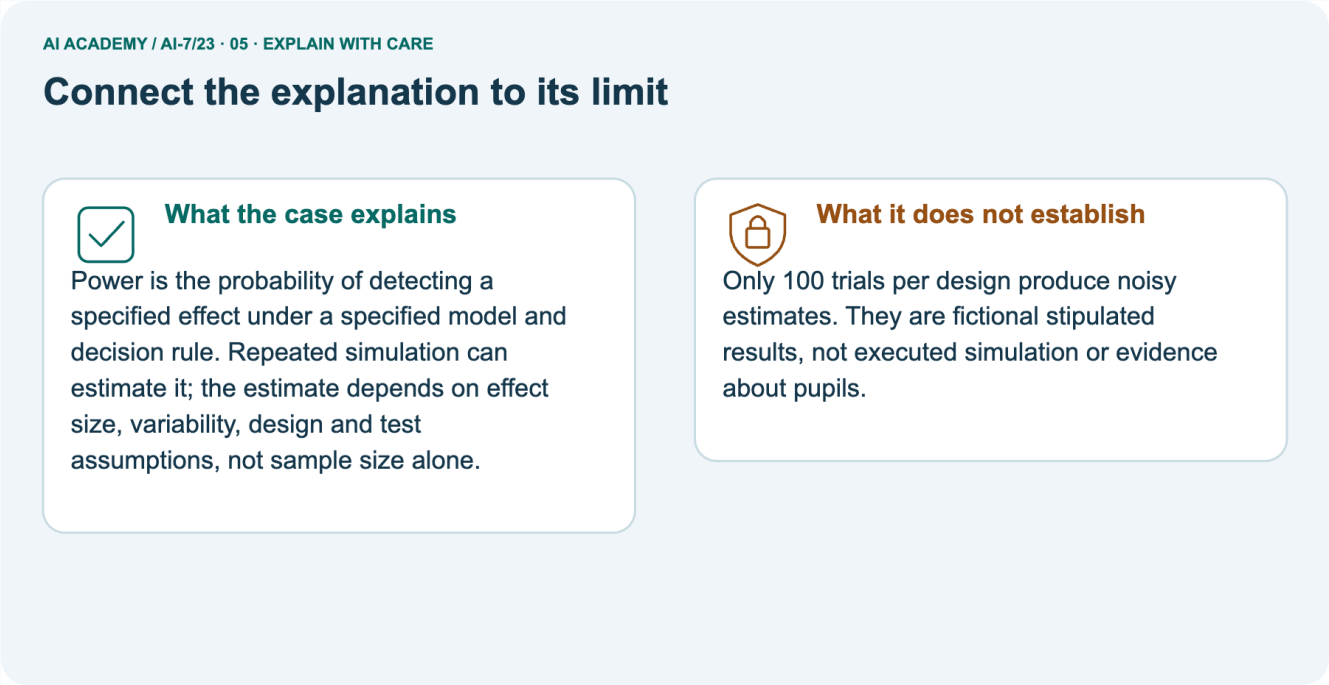
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Doubling cost adds 20 percentage points of simulated detection under these assumptions. Changed condition: Change the assumed effect or variability but keep the sample sizes.

Expected result: The detection fractions need new simulation; the resource costs stay 40 and 80. Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

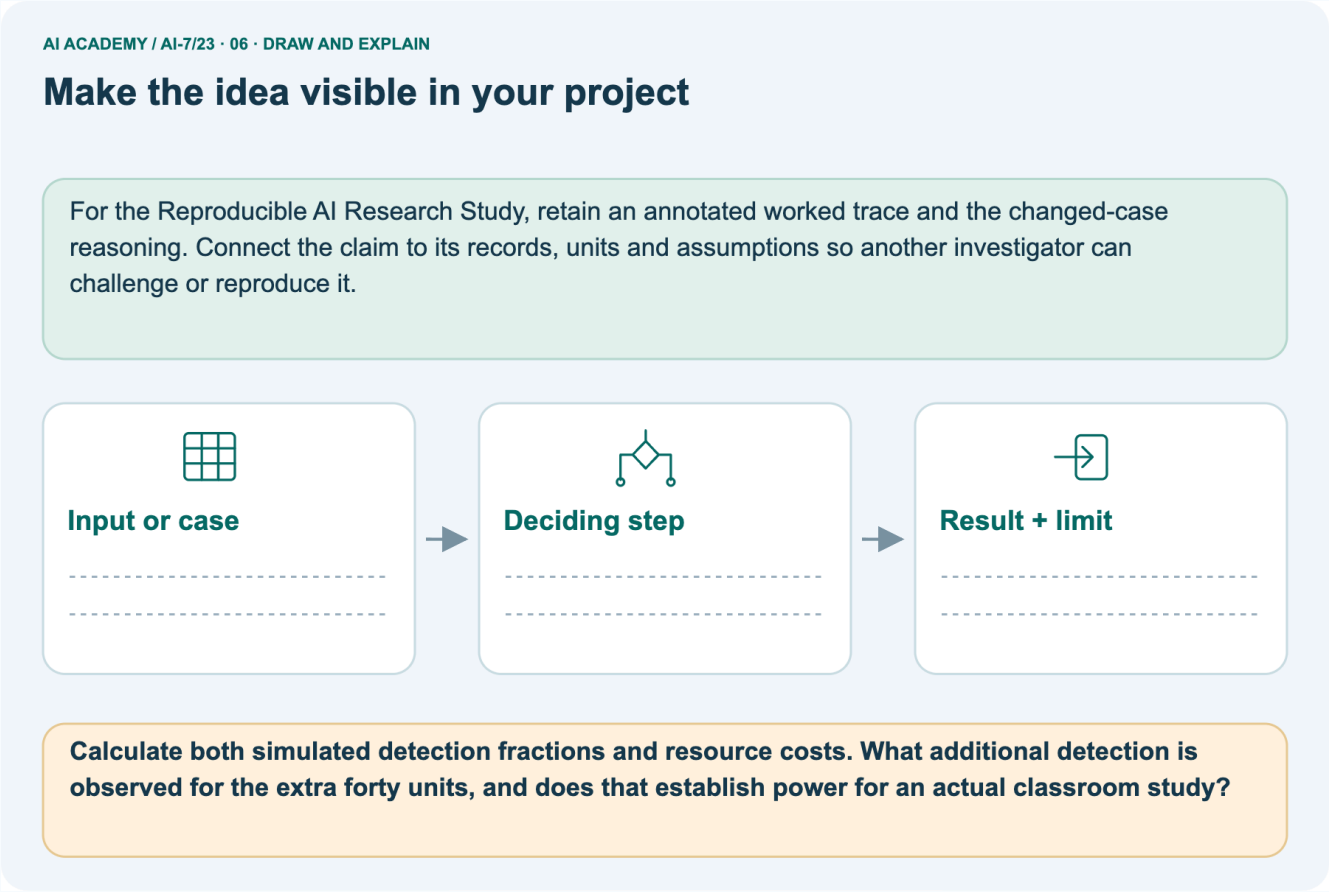
Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone. Limit: Only 100 trials per design produce noisy estimates.

They are fictional stipulated results, not executed simulation or **evidence** about pupils.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.